CNCA Centro Nacional de Computação Avançada

DEUCALION

Profiling @Deucalion

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Compiling the simple case study

Note: to use the score-p GUI start the ssh session with –X or –Y (`ssh –X or ssh -Y`)

Copy the project:

```
cd /projects/F202500010HPCVLABUMINHO/2026-2027cp/<username>   #quotaprojects
cp -r ../PLprojeto1/ .
```

Start a session in an ARM node:

```
srun --time=02:00:00 -p normal-arm  -A f202500010hpcvlabuminhoa -n 1 --x11 --pty bash
```

Load modules (e.g. GCC) , compile and run:

```
ml GCC
g++ -O3 MDseqClaudeAI.cpp
./a.out
```

Profiling with perf (sampling)

Flat time-profile:

```
perf record -F 1000./a.out  # 1000 samples per second
perf report –n # “-n” to display the number of samples
```

Call-graph time-profile:

```
perf record -g -F 1000./a.out  # 1000 samples per second
perf report –n # “-n” to display the number of samples
```

Use the “up” and “down” arrows and “E” to expand a particular entry

Profiling with Score-P (instrumentation)

Compile and run again with Score-P

```
ml Score-P
scorep g++ -O3 MDseqClaudeAI.cpp
./a.out  # or ./a.out 500 50
```

Load Cube:

```
ml CubeGUI
```

Read Cube file:

```
cd scorep-*
cube profile.cubex
```

Select time on Metric tree and verify hotspots in Call tree

